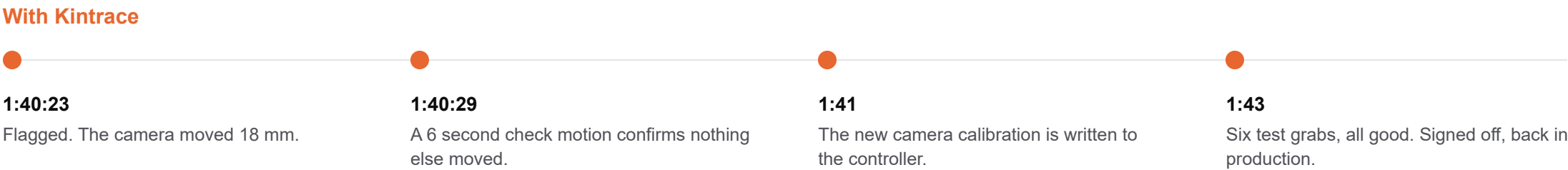
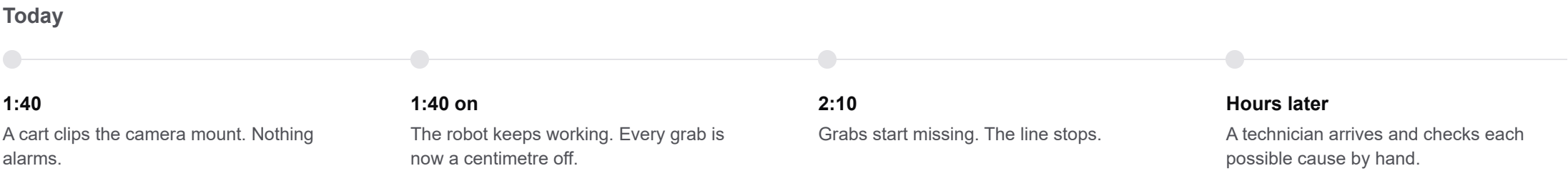


---

# Kintrace

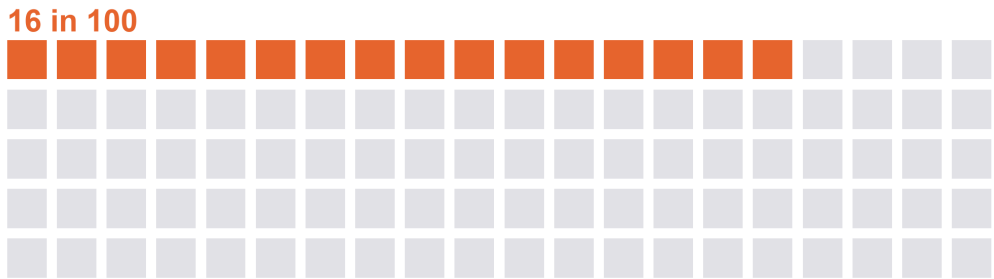
Robots get knocked out of place and nobody notices. Kintrace finds what moved and puts it back.

# At 1:40 a cart clips a camera. Nobody finds out until 2:10.



An illustration. The 18 mm and the timings are printed by the tool, running on our simulator. No cell like this exists yet.

You pay for it twice: in ruined training data, and in stopped lines.



**Robot training data**

An open audit of 4,959 recorded episodes flagged 15.6% of them for exclusion, and another half for review.

\$2.3M  
an hour

Up to, for an idle production line at a big car plant. Across the 500 largest manufacturers, unplanned downtime of every kind costs \$1.4 trillion a year, about 11% of revenue.

# Everything on the market tells you something is wrong, not what moved.

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## Monitoring dashboards

Ember, Foxglove, Formant

They watch the software and the logs. They will tell you a cell is slow or a run failed. They do not measure where the camera is, so they cannot name the physical cause.

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## Calibration hardware

Dynalog, CAPTRON, Renishaw

These do measure, and they are accurate. But you bolt a device into every cell you want covered, and each one mostly does the tool.

---

## A technician

How it is actually done

Someone drives out and rules out each cause by hand, one at a time, with the cell down while they do it. It works. It costs hours.

We have not found anyone doing all of it from software alone: when it changed, what moved, by how much, and the fix written back.

# Kintrace reads the robot's own logs, then says what moved and writes the fix.

\$ KINTRACE INCIDENT

RECORD.TXT

KINTRACE recommission record, Cell 3

=====

Changed at : 13:40:23 (lines up with protective stop at 13:40:23: collision detected, arm stopped)

seen in : camera vs arm at 13:40:23

seen in : camera vs table marker at 13:40:23

seen in : first missed pick at 13:40:25

What changed (6 s check motion)

- Camera moved: 18.1 mm, 0.04 deg (parts appear 18 mm off in the pick area)

Fix written to controller

- camera calibration updated

Verification (check motion re-run)

[ok ] Command timing 8 ms vs 8 ms normal

[ok ] Camera vs fixed cell marker table marker where it should be (1.0x noise)

[ok ] Camera vs joint encoders disagreement 1.0x sensor noise

[ok ] Fingertip vs known-good fingertip 0.1 mm from its known-good position

[ok ] Tool offset setting matches known-good

test picks: 6/6 ok

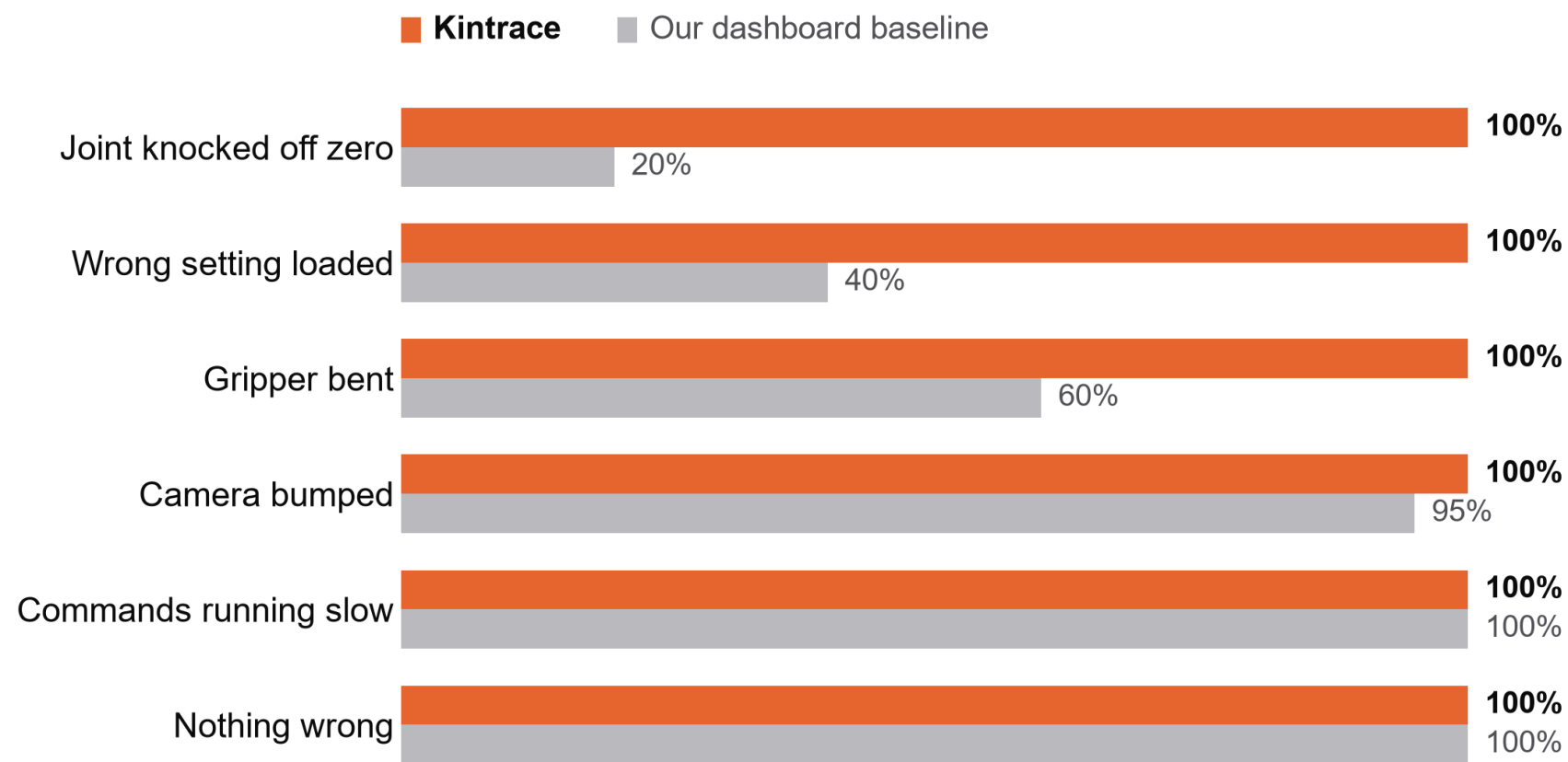
STATUS: GO



What we actually did: bump the camera, sized for about 20 mm at the fingertip.

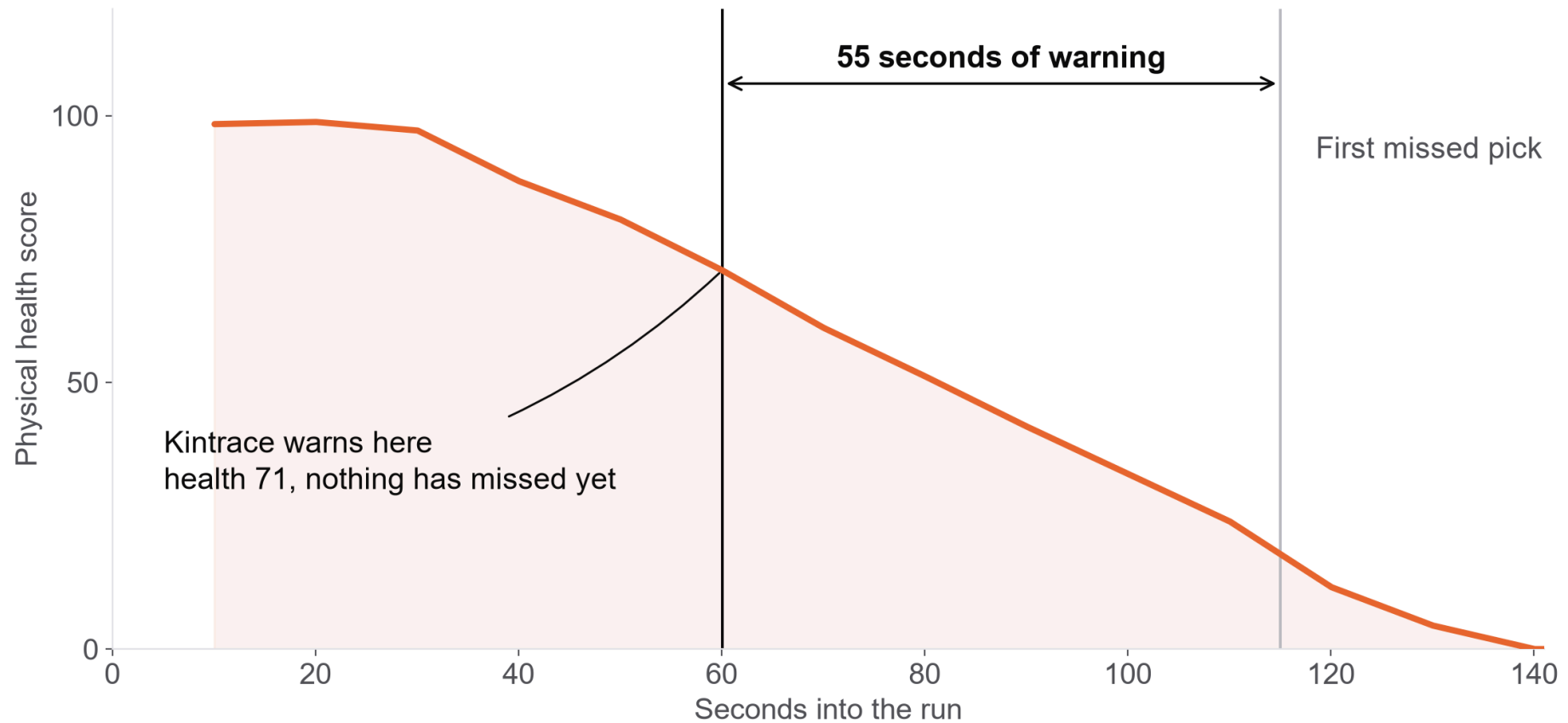
- **Notices**  
It finds the moment things changed and lines it up with the protective stop.
- **Explains**  
It names the cause and sizes it. Here: the camera, 18.1 mm.
- **Fixes**  
It writes the new calibration, re-checks, runs six test grabs, then signs GO or NO-GO.

In our simulator, Kintrace named the right cause every time. The dashboard baseline we wrote got 69%.



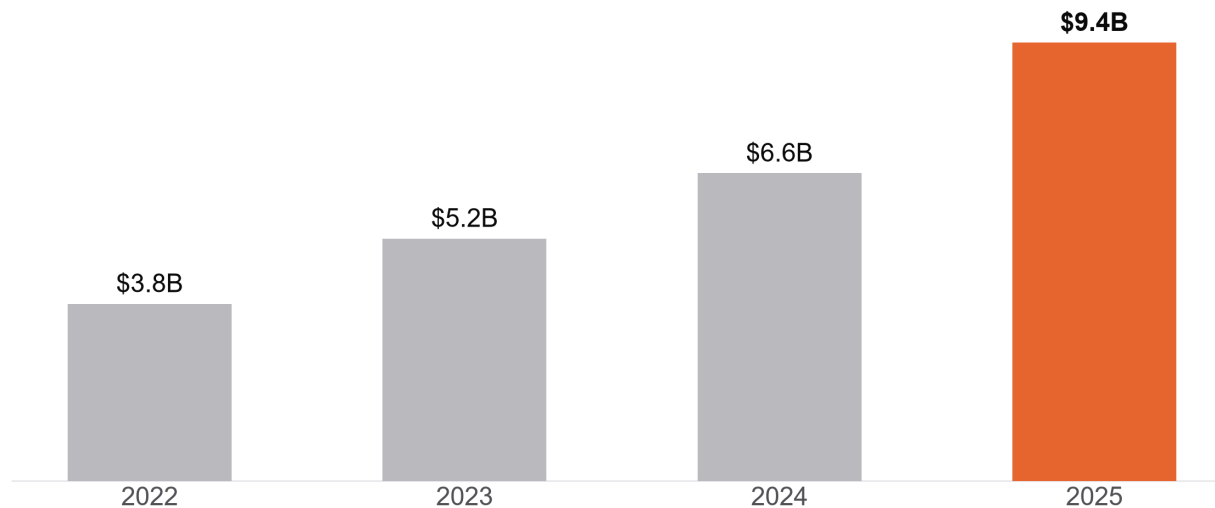
120 runs on our simulated UR5e pick cell, 20 per case, every fault sized to cause the same 20 mm miss. The baseline is ours, built from the signals a typical dashboard tracks. Never run on a real arm.

It also catches slow drift, about 55 seconds before the first grab misses.



One simulated run of a sagging camera mount. Across 20 drifts it warned before the first miss 19 times, named the cause right 20 times, and raised 0 false alarms in 230 healthy windows. Drift is sped up so a run takes minutes. We have not measured lead time at real-world drift rates.

There have never been more robots, and the money behind them doubled in three years.



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**5 million**

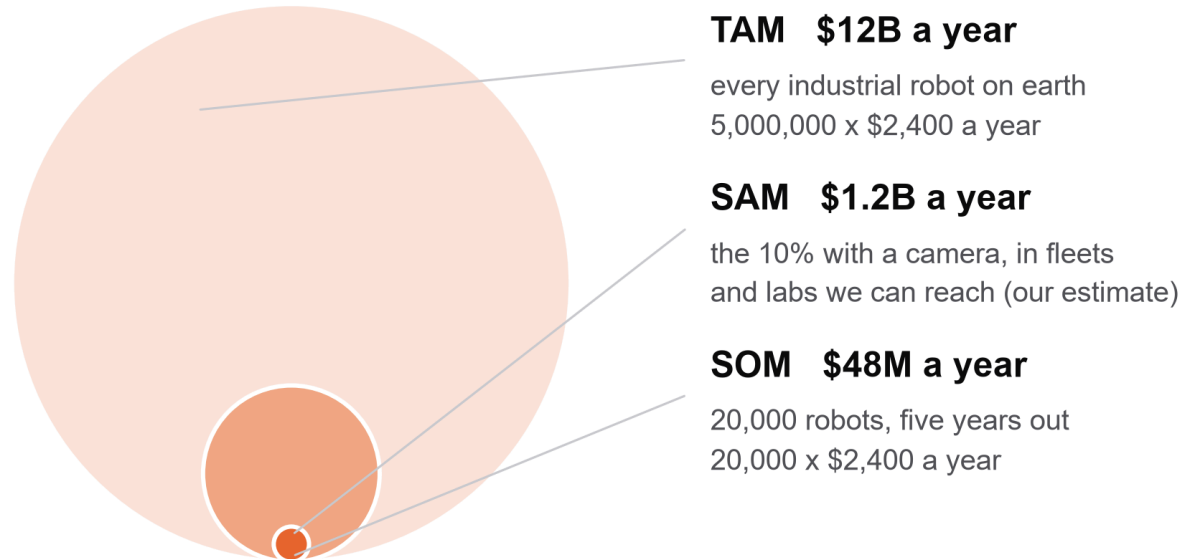
industrial robots now operating, a record, up 9% in a year. The US installed 38,500 and passed Japan. IFR, September 2026.

---

**47% a year**

forecast growth for physical AI, from \$0.89B in 2025 to \$15.2B by 2032. Every robot in it learns from recordings.

At \$200 a robot a month, the part we can realistically sell to is \$1.2B a year.



**Top down check:** \$22.5B was spent on industrial robot services in 2023, heading to \$41.6B by 2033. Our \$12B bottom up number sits inside a services market that is already twice its size.

# \$200 a robot a month, against a technician visit that costs \$1,500 to \$3,000.

## How we charge

**\$200**

per robot, per month, for always-on watching

**~\$300**

per incident we find and fix. Planned pilot pricing, no pilots yet

**\$1,500 to \$3,000**

what the technician visit it replaces costs today (our estimate)

## Who we sell to, in order

Robot learning teams first, they feel bad data fastest. Then robots-as-a-service fleets, who pay for every callout. Then robot makers.

Needs hardware

Finds and fixes the cause

### Dynalog, CAPTRON

Finds and fixes it, but you mount hardware in every cell.

### Kintrace

Finds it, sizes it, writes the fix, certifies the cell. Software only.

Software only

### Robot makers

Built-in mastering routines. One brand, run by a technician.

### Ember, Foxglove, dataset checkers

Show you something is off. No geometry, no fix.

Only shows a problem

# Raising [\$150K] to put Kintrace on real arms and into three paid pilots.

## Where it goes

- A desk arm rig, then time on a lab arm, to reproduce all five faults on real hardware.
- Three paid pilots with robot learning teams.
- A reader so fleets can send their own robot logs, in the formats they already record.

## Team

### Rithik Satarla

Physical AI research intern at CMU. Harvard Innovation Labs. Wrote all of the above.

[Cofounder]

## Next six months

- **Nov 2**  
YC W27 application in
- **Month 1-2**  
Five faults reproduced on the desk arm, numbers published
- **Month 3-4**  
Same five on a lab arm, and the first paid pilot
- **Month 5-6**  
Three paid pilots running, real log reader shipped